Systematic Evaluation of Highly Variable Gene Selection and Batch-Aware Approaches in Single-Cell RNA seq

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Background

Highly variable gene (HVG) selection is a key step in single-cell RNA-seq analysis, shaping which genes drive PCA, UMAP, neighbor graphs, and clustering. Different HVG methods, including Seurat, Seurat v3, scran-MV, and SCT proxy, employ fundamentally different variance models; however, their behavior is rarely evaluated side by side under matched conditions.

Objectives

Using two single-cell RNA seq: PBMC 10k (public) and CHD 30k (public) datasets,

- Benchmark four HVG selection methods on scRNA-seq datasets
- Assess gene-level similarity and biological biases
- Evaluate clustering performance as HVG count increases (K = 200–6000)
- Compare performance stability
- Determine how strongly each HVG strategy depends on dataset characteristics

Methods

All methods were reimplemented in Scanpy:

- Seurat: dispersion-based log-normalization
- Seurat v3: model-based decomposition of technical vs biological variance
- scran-MV: loess mean–variance trend residuals
- SCT proxy: Pearson residual variance from raw counts

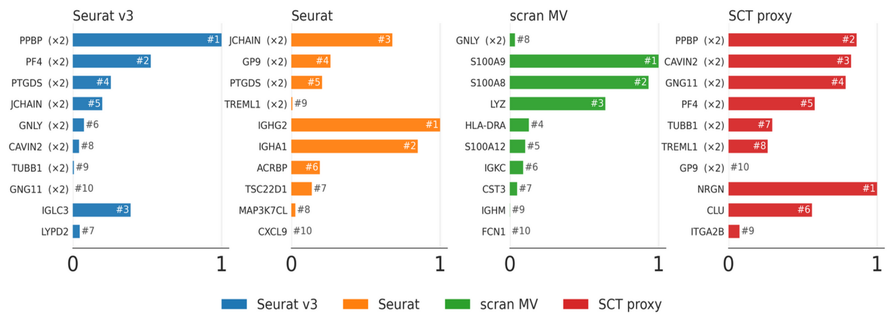
Downstream steps were identical across methods:

PCA → UMAP → Clustering.

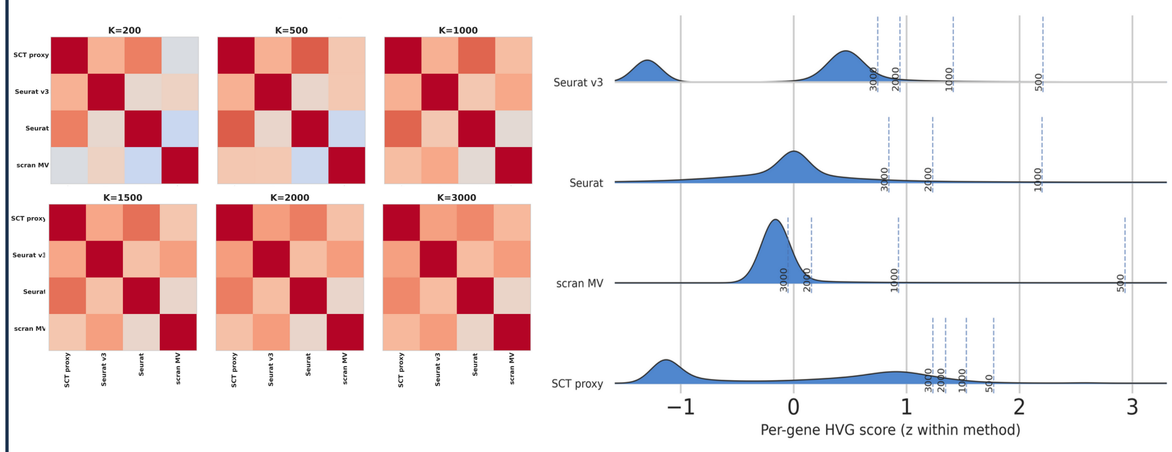
Metrics: ARI, NMI, kNN accuracy, ASW, Variance Ratio, invLISI, and ΔARI = change in ARI.

Results

Distinct Variance Signature: Different HVG methods show persistent, model-specific variance signatures that drive distinct gene-selection biases and prevent full concordance—even at large feature counts—ultimately leading to differing clustering.

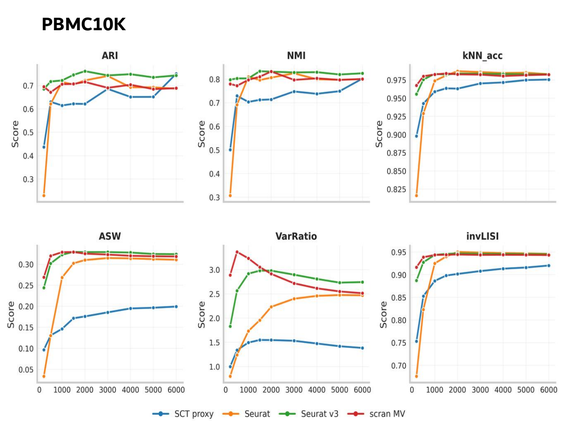


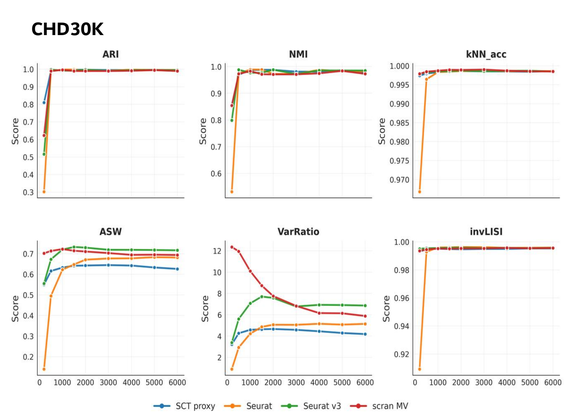
Variance Modeling: Variance distributions matched each method's selectionseparation, while Seurat and scran-MV applied smoother weighting. bias: Seurat v3 and the SCT proxy produced sharper high-variance

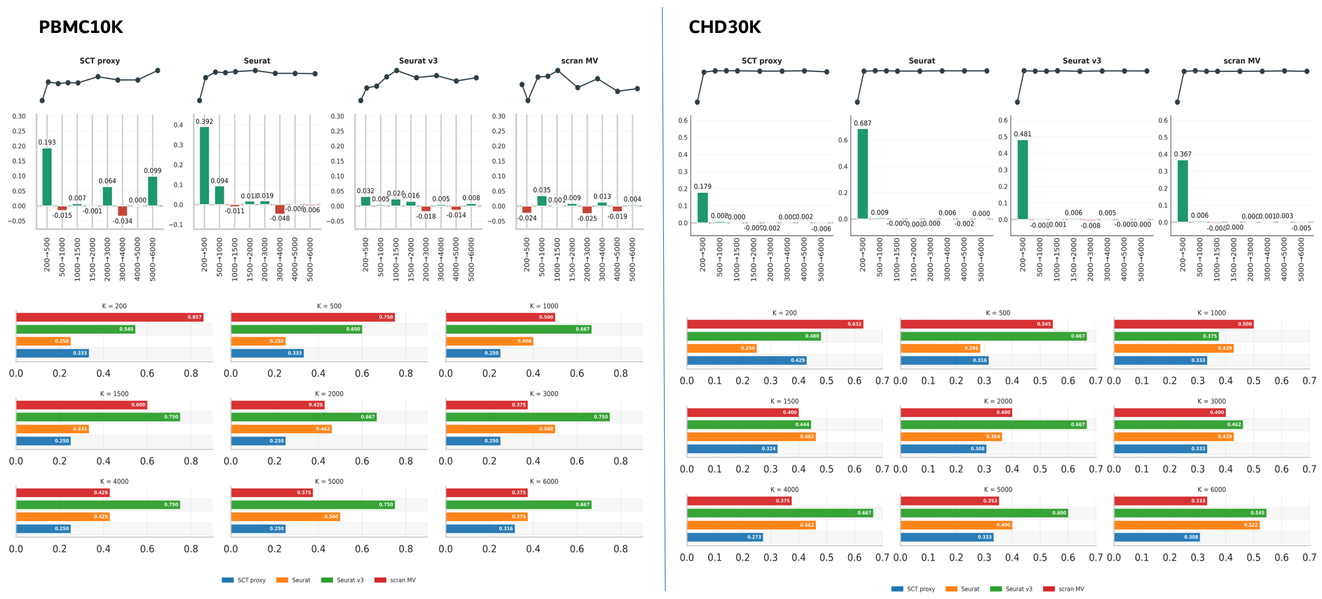


Clustering Performance: Dataset complexity dictated performance.

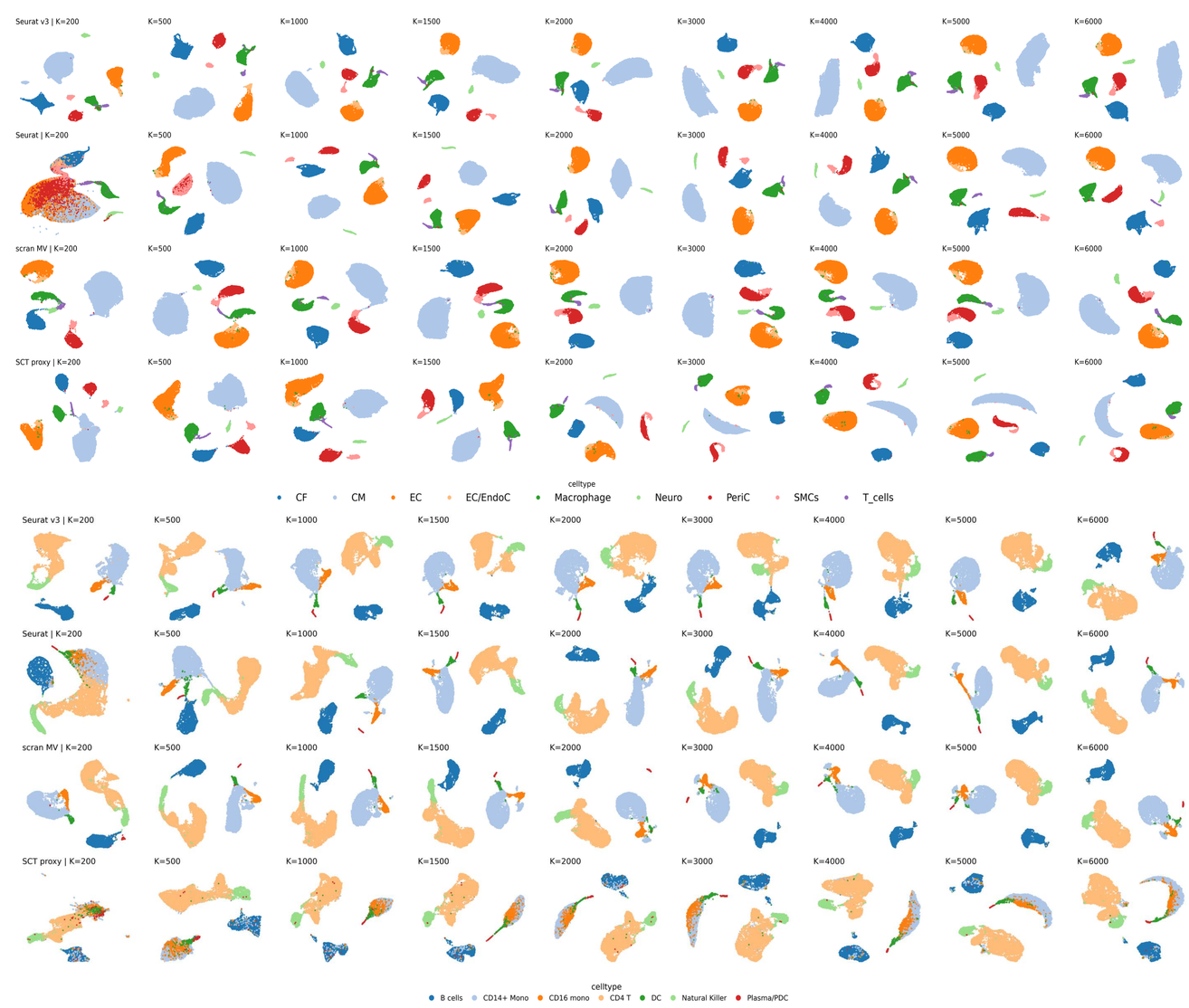
- PBMC (moderate complexity): Methods diverged; Seurat v3 was most stable across K, scran-MV performed best at low K, Seurat improved with more features, and SCT proxy performed worst.
- CHD (highly separable): All methods achieved near-perfect clustering once K ≥ 500, with minimal differences.



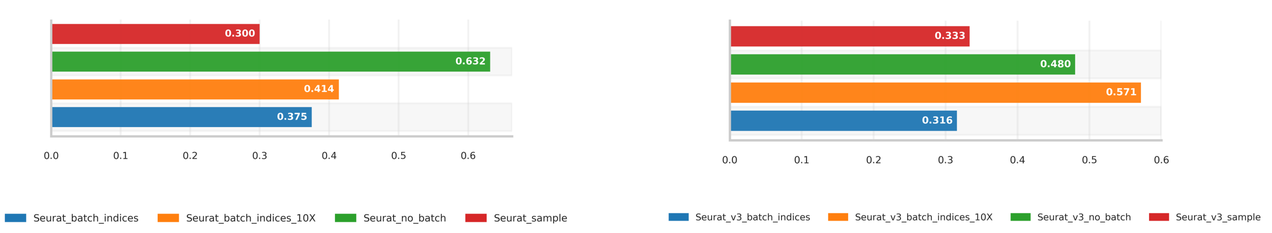




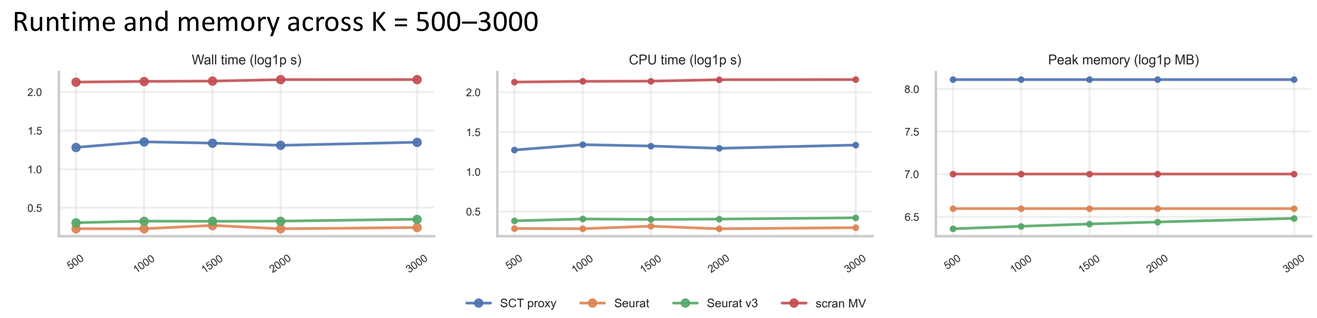
Dataset Speific Sensitivity: UMAPs across HVG methods and feature counts show method-dependent structure recovery in PBMC where SCT proxy lags versus rapid convergence in CHD, indicating HVG choice matters in low-signal data but is minimal in strongly separable datasets.



Performance Ranking: Inverse average performance rank comparing Seurat and Seurat v3 methods across all evaluation metrics.



Runtime and memory across K = 500–3000



Conclusion

HVG selection showed clear method-specific behavior, with performance depending on both the way each method models gene variability and the number of selected features. In complex data, HVG choice and K had a substantial impact, while well-separated structure led to convergence once K ≥ 500. Although larger HVG sets improved robustness, methods remained biased by their underlying variability–modeling assumptions. Overall, HVG strategy matters most in challenging datasets, with batch-aware effects to be explored further as more datasets are assessed.

References

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